

Breast Cancer Diagnosis Using an End-to-End Machine Learning Pipeline

CSTE 3207 — Machine Learning Project | Team Members: [Student 1], [Student 2], [Student 3]

Abstract

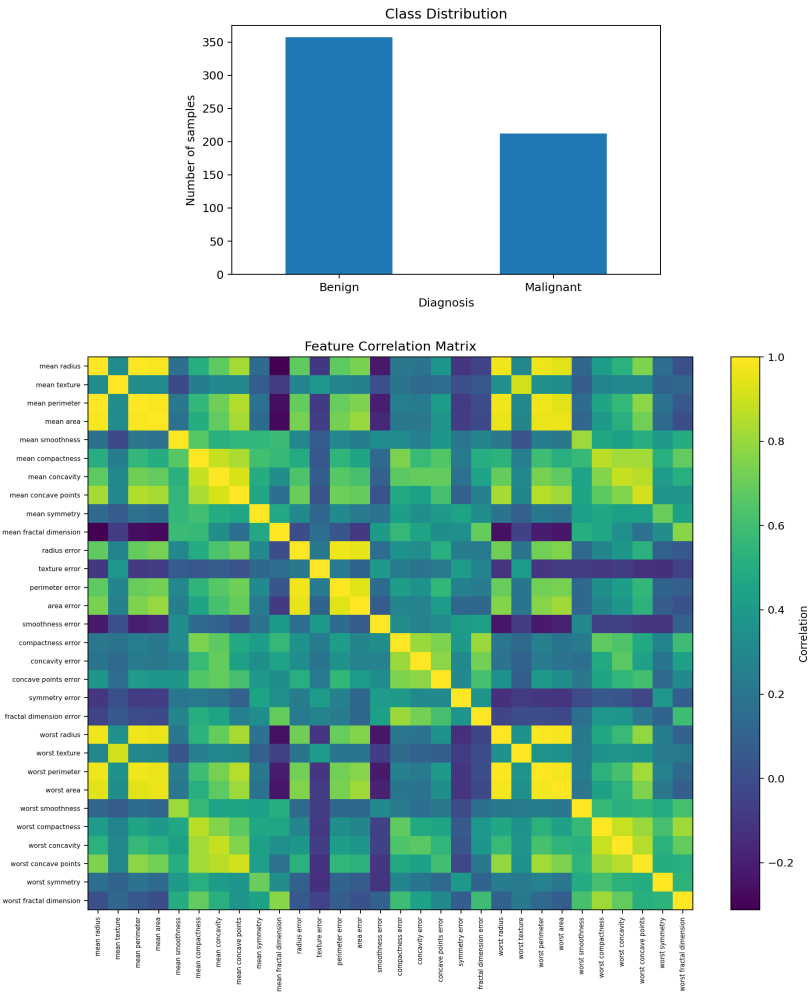
This project develops an end-to-end supervised machine learning pipeline for classifying breast tumor samples as malignant or benign. The study uses the Breast Cancer Wisconsin (Diagnostic) dataset with 569 samples and 30 numeric predictive features. The workflow includes problem formulation, exploratory data analysis, preprocessing, feature engineering, baseline and advanced modeling, hyperparameter tuning, cross-validation, quantitative evaluation, and error analysis. The best test-set model in this run was Logistic Regression, achieving accuracy 0.982, precision 1.000, recall 0.952, F1 0.976, and ROC-AUC 0.996. The model is intended for academic evaluation rather than clinical decision-making.

1. Introduction & Related Work

Breast tumor diagnosis is a binary classification problem in which a tumor can be categorized as malignant or benign. Machine learning can learn relationships between measurable characteristics of cell nuclei and the diagnosis label. The Wisconsin Diagnostic dataset is a standard benchmark for this task. Prior work has evaluated methods such as Support Vector Machines and neural networks; this project emphasizes the complete pipeline required by the course guideline rather than accuracy alone.

2. Dataset & Exploratory Data Analysis

The dataset contains 569 observations and 30 continuous predictive features. The target contains 212 malignant and 357 benign samples. Features describe radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension, with mean, standard-error, and worst-value groups. The original dataset contains 0 missing values. A leakage-safe median imputation step is nevertheless included for robustness. Outliers are inspected rather than automatically removed.



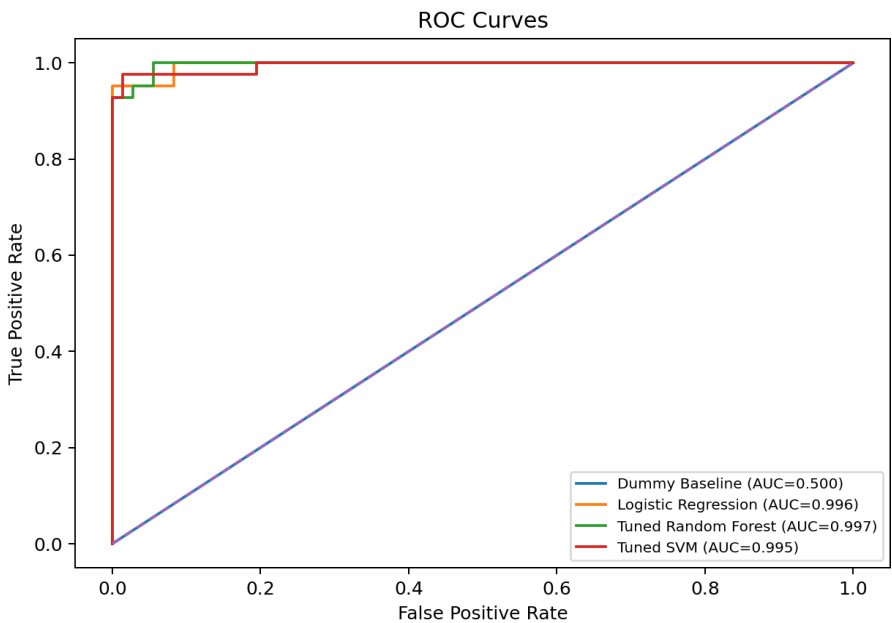
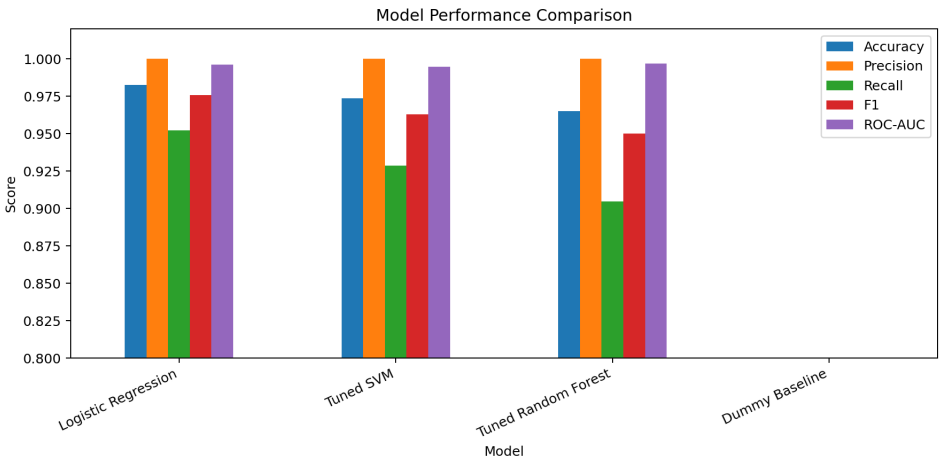
3. Methodology

The data are split into 80% training and 20% testing with stratification and random state 42. Three ratio-based features are engineered. A Dummy Classifier provides the baseline; Logistic Regression is a simple learning model; Random Forest and SVM are advanced candidates. StandardScaler is used for Logistic Regression and SVM. GridSearchCV with five-fold stratified cross-validation and F1 scoring tunes Random Forest and SVM. The best Random Forest parameters were {'model__max_depth': None, 'model__min_samples_split': 2, 'model__n_estimators': 400}; the best SVM parameters were {'model__C': 10, 'model__gamma': 'scale', 'model__kernel': 'rbf'}.

4. Experiments & Results

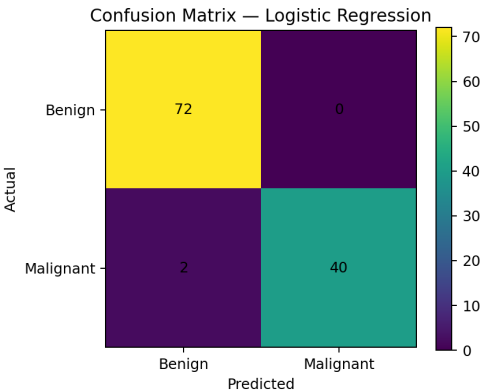
The models were evaluated using accuracy, precision, recall, F1-score and ROC-AUC. The held-out test set was not used during hyperparameter selection. The comparison is summarized in Table 1.

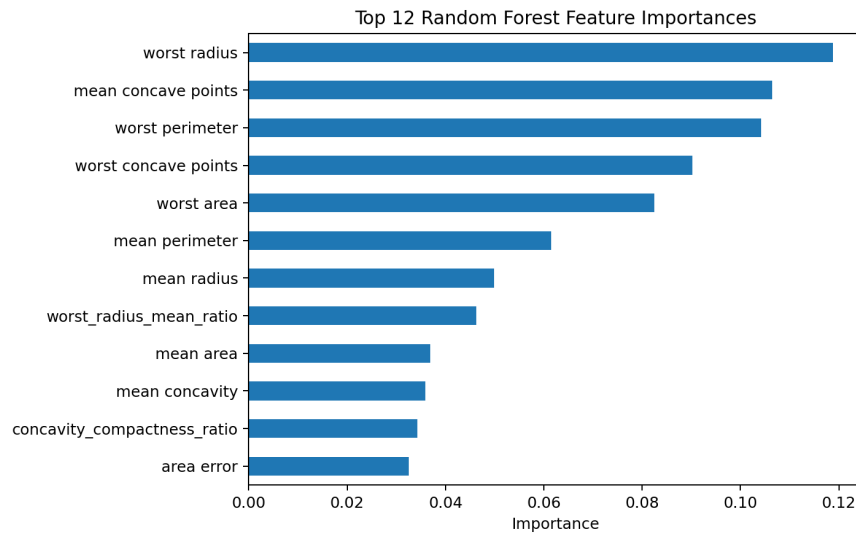
Model	Acc.	Prec.	Recall	F1	AUC
Logistic Regression	0.982	1.000	0.952	0.976	0.996
Tuned SVM	0.974	1.000	0.929	0.963	0.995
Tuned Random Forest	0.965	1.000	0.905	0.950	0.997
Dummy Baseline	0.632	0.000	0.000	0.000	0.500



5. Discussion & Error Analysis

The strongest test-set performance in this run came from Logistic Regression. The test set contained 2 misclassified cases for the best model. False negatives and false positives should both be examined, especially because the positive class represents malignant tumors. Feature importance provides model-specific evidence about predictive usefulness, not causality. Limitations include the relatively small benchmark dataset, possible sampling/measurement biases, and the need for independent clinical validation.





6. Conclusion & Future Work

The project completed the required end-to-end ML stages. The best model achieved $F1=0.976$ and $ROC-AUC=0.996$ on the held-out test set. Future work includes nested cross-validation, probability calibration, external validation, dimensionality reduction, class-weighted optimization, and comparison with gradient boosting or neural networks.

7. Author Contributions

Student 1: dataset preparation, EDA and preprocessing. Student 2: modeling, tuning and evaluation. Student 3: error analysis, report writing and presentation. Replace placeholders with actual names.

References

1. UCI Machine Learning Repository, “Breast Cancer Wisconsin (Diagnostic),” Dataset 17. 2. F. Pedregosa et al., “Scikit-learn: Machine Learning in Python,” JMLR, 2011. 3. A. F. Agarap, “On Breast Cancer Detection: An Application of Machine Learning Algorithms on the Wisconsin Diagnostic Dataset,” arXiv:1711.07831, 2017. 4. CSTE 3207 Project Guideline supplied for the course.